EXAMPLE: FOR A 2D INCOMPRESSIBLE, IRROTATIONAL FLOW, SUPERPOSITION OF A DOUBLET & UNIFORM FLOW AROUND A CIRCULAR CYLINDER. OBTAIN THE STREAMFUNCTION & VELOCITY POTENTIAL FOR THIS FLOW PATTERN. FIND THE VELOCITY FIELD, LOCATE THE STAGNATION POINTS.

$\psi = \psi_{DOUBLET} + \psi_{UNIFORM} = -\frac{\Lambda}{r}\sin\theta + U r\sin\theta$, $\phi = \phi_{DOUBLET} + \phi_{UNIFORM} = -\frac{\Lambda}{r}\cos\theta - Ur\cos\theta$

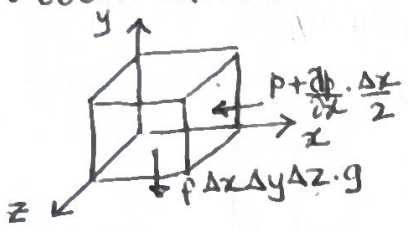
$u_r = -\frac{\partial\phi}{\partial r} = -\frac{\Lambda}{r^2}\cos\theta + U\cos\theta$, $u_\theta = -\frac{1}{r}\frac{\partial\phi}{\partial\theta} = -\frac{1}{r}\left[\frac{\Lambda}{r}\sin\theta + Ur\sin\theta\right] = -\frac{\Lambda}{r^2}\sin\theta - U\sin\theta$

STAGNATION PTS. $\Rightarrow \vec{v}=0 \Rightarrow -\frac{\Lambda}{r^2}\cos\theta + U\cos\theta = 0$, $-\frac{\Lambda}{r^2}\sin\theta - U\sin\theta = 0 \Rightarrow \cos\theta\left(U - \frac{\Lambda}{r^2}\right) = -\sin\theta\left(U + \frac{\Lambda}{r^2}\right) = 0$

THESE ARE SIMULTANEOUSLY SATISFIED IF $r = \sqrt{\Lambda/U}$, $\sin\theta = 0 \Rightarrow \theta = n\pi$. } FOR LYING ON CYLINDER SURFACE $a = \sqrt{\Lambda/U}$.

• HAVE TO TALK ABOUT STATE OF STRESS, SURFACE & BODY FORCES, PRESSURE & SHEAR
• DECIDING WHEN TO USE THE INVISCID ANALYSIS THAT WE ARE OUTLINING HERE IS DECIDED BY SCALING ANALYSIS, WE WILL COVER THAT IN DETAIL IN OUR DISCUSSION ON DIMENSIONAL ANALYSIS

• EULER EQUATIONS:



NET FORCE $= -\left(\frac{\partial p}{\partial x}\Delta x\hat{i} + \frac{\partial p}{\partial y}\Delta y\hat{j} + \frac{\partial p}{\partial z}\Delta z\hat{k}\right) - \rho g\,\Delta x\Delta y\Delta z(-\hat{j}) = \rho\cdot\frac{Dv}{Dt}\cdot\Delta x\Delta y\Delta z$

$\Rightarrow -\left(\frac{\partial p}{\partial x}\hat{i} + \frac{\partial p}{\partial y}\hat{j} + \frac{\partial p}{\partial z}\hat{k}\right) + \rho g(-\hat{j}) = \rho\frac{Dv}{Dt}$ } FINAL FORM OF EULER'S EQUATION

IMPORTANT ELEMENTS: CONVECTIVE ACCELERATION, NONLINEARITY

EXAMPLE: FOR A WATER FLOW IN A PIPE, DETERMINE THE PRESSURE GRADIENT REQUIRED TO ACCELERATE THE WATER AT 6 m/s² FOR (A) HORIZONTAL PIPE, (B) VERTICAL PIPE, WITH WATER FLOWING UPWARD, (C) VERTICAL PIPE WITH WATER FLOWING DOWNWARD. EXPLAIN & INTERPRET YOUR RESULTS.

EULER'S EQUATION $\Rightarrow \rho Dv/Dt = -\nabla p + \rho g$ $\Rightarrow$ HORIZONTAL PIPE $\Rightarrow \rho\cdot 6 = -\nabla p = -6\times 1000 = 1000$

VERTICAL PIPE $\Rightarrow \rho\cdot 6 = -\nabla p - \rho\cdot g \Rightarrow \nabla p = -\rho g - 6\rho = -15810$ (UPWARD)

VERTICAL PIPE $\Rightarrow -6\rho = -\nabla p - \rho g \Rightarrow \nabla p = 6\rho - \rho g = -3810$ (DOWNWARD)

PRESSURE GRADIENT IS MUCH SMALLER FOR DOWNWARD MOTION AS GRAVITY SUPPORTS THE ACCELERATION!!

• EULER EQUATIONS ALONG STREAMLINE COORDINATES.

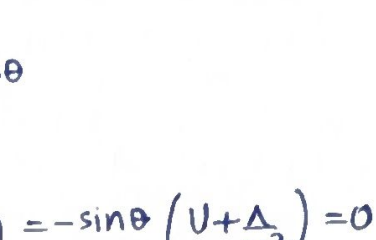
$\left[p - \frac{\partial p}{\partial s}\cdot\frac{ds}{2}\right]dn\,dx - \left[p + \frac{\partial p}{\partial s}\cdot\frac{ds}{2}\right]dn\,dx - \rho g_s\,ds\,dn\,dx = \rho a_s\cdot ds\,dn\,dx \Rightarrow -\frac{\partial p}{\partial s} - \rho g_s = \rho a_s$ (STREAMLINE DIRECTION)

$\Rightarrow -\frac{1}{\rho}\frac{\partial p}{\partial s} - g_s = \frac{\partial v}{\partial t} + v\frac{\partial v}{\partial s}$ (FLOW IS ONLY ALONG STREAMLINE), $-\frac{1}{\rho}\frac{\partial p}{\partial s} - g_s = v\frac{\partial v}{\partial s}$ (STEADY STATE) → FOR $g_s = 0$, A DECREASE IN VELOCITY IS ACCOMPANIED BY INCREASE IN PRESSURE

ALONG NORMAL COORDINATES, $\left[p - \frac{\partial p}{\partial n}\cdot\frac{dn}{2}\right]ds\,dx - \left[p + \frac{\partial p}{\partial n}\cdot\frac{dn}{2}\right]ds\,dx - \rho g_n\,ds\,dn\,dx = \rho a_n\,ds\,dn\,dx \Rightarrow -\frac{1}{\rho}\frac{\partial p}{\partial n} - g_n = a_n$

$\Rightarrow -\frac{1}{\rho}\frac{\partial p}{\partial n} - g_n = \frac{-v^2}{R}$ (NO FLOW ALONG NORMAL DIRN, CENTRIPETAL ACCN) → PRESSURE INCREASES IN THE DIRECTION OUTWARD FROM THE CENTER OF CURVATURE OF THE STREAMLINES.

EXAMPLE: AIR FLOWS IN A 2D BEND OF WIDTH W IN A DUCT AS SHOWN IN THE FIGURE. THE VELOCITY PROFILE IS SIMILAR TO A FREE VORTEX PROFILE GIVEN BY $v_\theta = C/r$. SHOW THAT THE FLOW RATE Q CAN BE EXPRESSED IN THE FORM $Q = K\sqrt{\Delta p}$, WHERE K IS A CONSTANT. (NO BODY FORCES)



NORMAL TO STREAMLINES, $\frac{1}{\rho}\partial p/\partial n = v^2/R \Rightarrow \frac{1}{\rho}\partial p/\partial n = C^2/r^2/r = C^2/r^3 \Rightarrow \partial p/\partial n = \rho C^2/r^3$

$\Rightarrow p_2 - p_1 = \frac{\rho C^2}{(-2)}\left(\frac{1}{r^2}\right)\Big|_{r_1}^{r_2} = \frac{\rho C^2}{2}\left[\frac{1}{r_1^2} - \frac{1}{r_2^2}\right] = \Delta p$

FLOW RATE, $Q = \int v\cdot dA = \int_{r_1}^{r_2} \frac{C}{r}\cdot dr\cdot W$ (WIDTH INTO PLATE) $= WC\ln(r_2/r_1) \Rightarrow C = \frac{Q}{W\cdot\ln(r_2/r_1)}$

$\Rightarrow \frac{\rho}{2}\cdot\frac{Q^2}{W^2\ln^2(r_2/r_1)}\left[\frac{1}{r_1^2} - \frac{1}{r_2^2}\right] = \Delta p$

$\Rightarrow Q^2 = \frac{2W^2}{\rho}\ln^2(r_2/r_1)\cdot\frac{r_1^2 r_2^2}{r_2^2 - r_1^2}$

(BERNOULLI DERIVATION ALONG STREAMLINE)

• ALONG STREAMLINE COORDINATES, $-1/\rho\,\partial p/\partial s - g\partial z/\partial s = v\,\partial v/\partial s \Rightarrow -\frac{1}{\rho}\frac{\partial p}{\partial s}\cdot ds = g\frac{\partial z}{\partial s}\cdot ds + v\frac{\partial v}{\partial s}\cdot ds$

$\Rightarrow -\frac{1}{\rho}\cdot dp|_{ALONG S} = g\,dz|_{ALONG S} + v\cdot dv|_{ALONG S} \Rightarrow dp/\rho + v\,dv + g\,dz = 0\,|_{ALONG S} \Rightarrow p/\rho + v^2/2 + gz = CONSTT.$

STEADY, INCOMPRESSIBLE, INVISCID, ALONG A STREAMLINE. (IMPORTANT RESTRICTIONS)

• ALONG RECTANGULAR COORDINATES, $v\cdot\nabla v = -\frac{1}{\rho}\nabla p - g\hat{k} \Rightarrow$ MULTIPLYING BY ds, $v\cdot\nabla v\cdot ds = -\frac{1}{\rho}\nabla p\cdot ds - g\hat{k}\cdot ds$

FROM VECTOR CALCULUS, $v\cdot\nabla v = \frac{1}{2}\nabla(v\cdot v) - v\times(\nabla\times v)$

FOR IRROTATIONAL, $\nabla\times v = 0 \Rightarrow \frac{1}{2}\nabla(v\cdot v) = -\nabla(p/\rho) - g\hat{k} \Rightarrow \frac{1}{2}\nabla(v\cdot v) = -\nabla(p/\rho) - \nabla(gz)$

ALONG AN ARBITRARY $dr = dx\hat{i} + dy\hat{j} + dz\hat{k}$, $\frac{1}{2}\nabla(v\cdot v)\cdot dr = -\nabla(p/\rho)\cdot dr - \nabla(gz)\cdot dr$

$\Rightarrow d(v^2/2) + d(p/\rho) + d(gz) = 0 \Rightarrow$ BERNOULLI.

STEADY, INCOMPRESSIBLE, INVISCID, IRROTATIONAL (ALONG A STREAMLINE IS NOT NECESSARY)

• WHEN ROTATIONAL, $v\cdot\nabla v\cdot ds = \frac{1}{2}\nabla(v\cdot v)\cdot ds - v\times(\nabla\times v)\cdot ds = \frac{1}{2}\nabla(v\cdot v)\cdot ds + (\nabla\times v)\times v\cdot ds$ (WILL BE ZERO AS $v \parallel ds$)

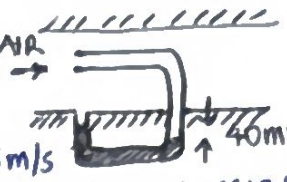
$\nabla p\cdot ds = \frac{\partial p}{\partial x}dx + \frac{\partial p}{\partial y}dy + \frac{\partial p}{\partial z}dz\,|_{ALONG STREAMLINE} = dp\,|_{ALONG STREAMLINE}$.

$-g\hat{k}\,ds = -g\,dz\,|_{ALONG STREAMLINE}$.

ADDING, $d(v^2/2) + dp + d(gz) = 0\,|_{ALONG STREAMLINE}$. } ALTERNATIVE DERIVATION ALONG A STREAMLINE.

• WALL PRESSURE TAPS, ITS IMPORTANT TO MENTION/RECALL THAT WHEN STREAMLINES ARE PARALLEL THERE IS NO PRESSURE VARIATION PERPENDICULAR TO THE STREAMLINES.

EXAMPLE: A PITOT TUBE IS INSERTED IN AN AIR FLOW TO MEASURE THE FLOW SPEED. AIR
THE STATIC PRESSURE IS MEASURED AT THE SAME LOCATION USING A WALL PRESSURE TAP.
IF THE PRESSURE DIFFERENCE IS 40 mm Hg, DETERMINE THE FLOW SPEED.



$P_0 = P + \frac{1}{2}\rho V^2 \Rightarrow (2 \times 40\,\text{mm Hg})/\rho_{AIR} = V^2 \Rightarrow V = \sqrt{\frac{2 \times 1000 \times 9.81 \times \frac{40}{1000} \times 13.6}{1.23}} = 93.15\,\text{m/s}$

(IS THIS INCOMPRESSIBLE??)

EXAMPLE: STEADY, FRICTIONLESS & INCOMPRESSIBLE FLOW FROM LEFT TO RIGHT OVER A STATIONARY CIRCULAR CYLINDER OF RADIUS a IS REPRESENTED BY THE VELOCITY FIELD $V = U\left[1 - \left(\frac{a}{r}\right)^2\right]\cos\theta\,\hat{e}_r - U\left[1 + \left(\frac{a}{r}\right)^2\right]\sin\theta\,\hat{e}_\theta$

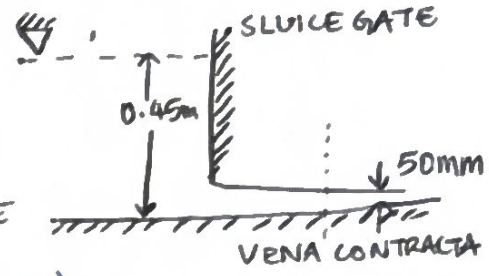
OBTAIN AN EXPRESSION FOR THE PRESSURE DISTRIBUTION ALONG THE STREAMLINE ON THE CYLINDER SURFACE $r=a$. DETERMINE THE LOCATIONS WHERE STATIC PRESSURE EQUALS FREESTREAM STATIC PRESSURE

ON SURFACE, $V = 0\,\hat{r} - 2U\sin\theta\,\hat{e}_\theta = -2U\sin\theta\,\hat{e}_\theta \Rightarrow P + \frac{1}{2}\rho V^2 = \text{CONST.} \Rightarrow P + \rho U^2\sin^2\theta = P_0$

FREESTREAM CONDITIONS, $P_\infty + \frac{1}{2}\rho U^2 = P_0 = P + 4U^2\sin^2\theta \Rightarrow P = P_\infty + \frac{1}{2}\rho U^2[1 - 4\sin^2\theta]$

FOR SAME PRESSURE, $1 - 4\sin^2\theta = 0 \Rightarrow \sin\theta = \pm\frac{1}{2} \Rightarrow \theta = \pi/6, \pi - \pi/6, \pi + \pi/6, 2\pi - \pi/6$.

EXAMPLE: WATER FLOWS UNDER A SLUICE GATE ON A HORIZONTAL BED AT THE INLET TO A FLUME. UPSTREAM FROM THE GATE, THE WATER DEPTH IS 0.45 m AND THE SPEED IS NEGLIGIBLE. AT THE VENA CONTRACTA DOWNSTREAM FROM THE GATE, THE FLOW STREAMLINES ARE STRAIGHT AND THE DEPTH IS 50 mm. DETERMINE THE FLOW SPEED DOWNSTREAM FROM THE GATE AND THE DISCHARGE IN CUBIC FEET PER SECOND PER FOOT OF WIDTH.



STEADY, INCOMPRESSIBLE, IF INVISCID THEN WE CAN APPLY BERNOULLI (ASSUMPTION)

CONSIDER STREAMLINE ALONG BOTTOM $\Rightarrow (p_{atm} + \rho g \times 0.45)/\rho + V_1^2/2 = (p_{atm} + \rho g \times 0.05/\rho) + V_2^2/2$

$\Rightarrow V_2 = \sqrt{2g \times 0.4} = 2.8\,\text{m/s} \Rightarrow Q = 2.8\,\text{m/s} \times 50\,\text{mm} = 0.14\,\text{m}^3/\text{s/width}$

EXAMPLE: A SMOOTHLY CONTOURED NOZZLE, WITH OUTLET DIAMETER $d = 20$ mm IS COUPLED TO A STRAIGHT PIPE USING FLANGES. WATER FLOWS IN THE PIPE OF DIAMETER 50 mm, AND THE NOZZLE DISCHARGES INTO THE ATMOSPHERE. FOR STEADY FLOW & NEGLECTING THE EFFECTS OF VISCOSITY, FIND THE VOLUME FLOW RATE IN THE PIPE CORRESPONDING TO A CALCULATED AXIAL FORCE OF 45.5 N NEEDED TO KEEP THE NOZZLE ATTACHED TO THE PIPE.

STEADY, INCOMPRESSIBLE, INVISCID, FLOW ALONG CENTER STREAMLINE $\Rightarrow p_1 + \rho V_1^2/2 = p_2 + \rho V_2^2/2$

CONSERVATION OF MASS, $A_1V_1 = A_2V_2 \Rightarrow D^2 \cdot V_1 = d^2 V_2 \Rightarrow V_2 = (D/d)^2 V_1 \Rightarrow p_1 + \rho V_1^2/2 = p_2 + \rho V_1^2 \cdot \left(\frac{D}{d}\right)^4 \cdot 1/2$

HAVE TO TALK ABOUT CONTROL VOLUME ANALYSIS BEFORE PROCEEDING WITH THIS PROBLEM!! $\Rightarrow p_1 - p_2 = \rho V_1^2/2\left\{\left(\frac{D}{d}\right)^4 - 1\right\}$

EXAMPLE: CONSIDER FRICTIONLESS, INCOMPRESSIBLE FLOW OF AIR OVER THE WING OF AN AIRPLANE FLYING AT 250 km/hr. THE AIR APPROACHING THE WING IS AT 78 kPa AND 13°C. AT A CERTAIN POINT IN THE FLOW, THE PRESSURE IS 75 kPa. CALCULATE THE SPEED OF AIR RELATIVE TO WING & AIR SPEED.

INCOMPRESSIBLE (VALID FOR AIR AT SMALL MACH NO.), STEADY, INVISCID, ALONG STREAMLINE.

$P_1 = 78\,\text{kPa}$, $T_1 = 13°C \Rightarrow \rho_1 = P_1/RT = 78 \times 1000/286.9 \times 286 = 0.95\,\text{kg/m}^3$ (INCOMPRESSIBLE, SO THIS MAY BE ASSUMED CONSTANT)

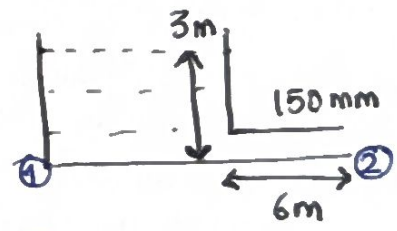
BERNOULLI $\Rightarrow p_2 + \rho g h_1 + \rho V_1^2/2 = p_2 + \rho g h_2 + \rho V_2^2/2 \Rightarrow V_2 = \sqrt{\frac{2}{\rho}(p_1 - p_2) + V_1^2} = \sqrt{\frac{2}{0.95}(78000 - 75000) + \left(\frac{250 \times 1000}{3600}\right)^2}$

↑ MEASURED RELATIVE TO AIRPLANE WING.

• EULER EQN. $\Rightarrow Dv/Dt = -\frac{1}{\rho}\nabla p - g\hat{k} \Rightarrow Dv/Dt \cdot ds = -\frac{1}{\rho}\nabla p \cdot ds - g\hat{k} \cdot ds \Rightarrow \frac{\partial v}{\partial t} ds + v\frac{\partial v}{\partial s} ds = -\frac{1}{\rho}\nabla p \cdot ds - g\hat{k} \cdot ds$

ADDITIONAL TERM IN THE TIME VARYING FORM OF THE BERNOULLI EQN. $\Rightarrow \frac{\partial v}{\partial t} \cdot ds + v\,dv + \frac{1}{\rho}dp + g\,dz = 0$

EXAMPLE: A LONG PIPE IS CONNECTED TO A LARGE RESERVOIR THAT IS INITIALLY FILLED WITH WATER UPTO A DEPTH OF 3m. THE PIPE IS 150 mm IN DIAMETER AND 6m LONG. DETERMINE THE FLOW VELOCITY LEAVING THE PIPE AS A FUNCTION OF TIME AFTER A CAP IS REMOVED FROM ITS FREE END.



INCOMPRESSIBLE, INVISCID, FLOW ALONG BOTTOM STREAMLINE

$\Rightarrow \cancel{p_1}^{\to p_{atm}} + \rho g \cdot 3m + \cancel{V_1^2/2}^{\to 0} = \int_1^2 \frac{\partial v}{\partial t} \cdot ds = \cancel{p_2}^{\to p_{atm}} + V_2^2/2 \Rightarrow V_2^2/2 + L \cdot dV_2/dt = g \cdot 3$

$\Rightarrow \int_0^{V_2} \frac{dV_2}{2 \cdot 3 \cdot g - V_2^2} = \int_0^t \frac{dt}{2 \times 6m}$

$\Rightarrow V_2 = \sqrt{2gh}\,\tanh\left[\frac{1}{2L}\sqrt{2gh}\right]$